Tanvi Muppala

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Education

The University of Texas at Austin — Austin, TX
B.S. Computer Science & Mathematics, Minor in Business — *Expected May 2028*
- Relevant Coursework: Data Structures, Algorithms and Complexity, Operating Systems, Computer Architecture, CS Pitch to Prototype, Linear Algebra, Probability & Statistics, Discrete Mathematics

Experience

Software Development Engineering Intern — May 2026 – Aug 2026
IBM Infrastructure — Power Systems Team — *Austin, TX*
- Built an end-to-end agentic AI system in Python (LangGraph, Ollama, A2A) with a Streamlit UI, deploying specialized agents for threat modeling, fix evaluation, and verification.
- Implemented an automated multi-agent feedback loop to analyze codebases for security threats, evaluate mitigations, and generate validated vulnerability fixes.
- Automated data-ingestion with MCP servers for NASA at IBM, accelerating research data retrieval and processing

Cybersecurity Analyst — Sep 2025 – Present
UT Regional Security Operations Center — *Austin, TX*
- Completed Hack The Box modules and co-authored Sherlock reports identifying threats and attack patterns

Undergraduate Course Assistant — Aug 2025 – Present
UTCS Think Lab — *Austin, TX*
- Scaled support from 1 to 3 sections totaling 60+ students weekly while assisting 250+ students in large lectures; deliver presentations, conduct office hours, and manage course content on Canvas

Software Engineering Intern — Jun 2025 – Sep 2025
MAG CP — *Dallas, TX*
- Automated cash-flow forecasting and reporting for 20+ properties with Python and Streamlit, cutting analysis time 70% and improving accuracy 40%; built investor email campaigns achieving 46% open rate

Product & Tech Lead — Oct 2024 – Aug 2025
Crsh Market — *Austin, TX*
- Led development of double-date mode and live ranking features on AWS Lambda with Twilio 2FA; organized events and photography while supporting 27 sold-out live shows generating over $850k in total revenue

Computational Researcher — Jan 2024 – May 2024
Freshman Research Initiative — *Austin, TX*
- Optimized Bader analysis code, reducing runtime from 5 minutes to under 10 seconds; authored solo research paper

Projects

Pintos Operating System | *C, Operating Systems* — Spring 2026
- Worked on a team to extend a toy OS with priority scheduling, argument passing, system calls for user programs, virtual memory, and a multi-threaded multi-level indexed file system

ClassScheduler | *C#, Blazor* — 2026
- Built a constraint-based scheduling engine that assigns UT Austin courses across 246 real classrooms based on previous classroom data and professor preferences

CrossTalk | *Python, JavaScript* — 2026
- Built a tool that scores 2–10 public repositories on composability vs. redundancy from their READMEs, surfacing reusable code for new projects

Technical Skills

Languages: Python, Java, C++, C#, JavaScript, HTML/CSS
Frameworks & Tools: LangGraph, Ollama, Streamlit, React Native, AWS Lambda, Git, Linux, Twilio, Oracle
Other: Agentic AI Workflows, MCP Servers, Security Threat Modeling, Adobe Suite, AWS Certification (In Progress)

Leadership & Awards

1st Place in Track / 3rd Overall among 500+ interns, National U.S. Intern Bobathon — Built Intern Helper Mode for onboarding timeline blueprints and company acronym translations
6th Place, Austin IBM Bobathon — RAG style MCP-powered document ingestion tool to parse/ index large PDFs
7th Place, Sandia National Labs TracerFire — investigated APT attacks as incident responder, tracking adversary movements and presenting mitigations to executives
Vice President, Dogs Without Borders — Organized fundraisers for animal welfare at local Mueller farmers markets
Social Officer, Indian Cultural Association — Organized community events and raised and managed funds